



APSCO

LINKU Project

CAMERA SYSTEM WORKING DOCUMENT

"ROTATION AND ROLLING-SHUTTER CORRECTION EXPERIMENT"

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Rotation and Rolling-Shutter Correction Experiment

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Abstract

This working document defines the experimental plan to characterize and mitigate rolling-shutter image distortion caused by spacecraft rotation during tether imaging. The LINKU stereo vision system uses rolling-shutter CMOS sensors (Sony IMX477), which read the image sensor row by row rather than capturing the full frame instantaneously. If the spacecraft rotates during that readout window, the image is geometrically sheared, an effect that is proportionally more severe for this mission than for typical photography because the tether itself occupies only one to three pixels of width, so even a small geometric shear is comparable in magnitude to the target being measured.

A first-pass calculation, using the camera system's own operating parameters (8 mm focal length, 1920x1440 capture resolution) and the IMX477 datasheet's stated frame-rate capability, indicates a tolerable rotation rate of approximately 1.05 degree per second under the software's current conservative 15 fps configuration, or approximately 2.76 degree per second if the sensor's faster full-resolution readout modes (up to 40 to 60 fps) are used instead, for a 3-pixel shear limit. This document plans the experiment to test both estimates directly and to validate a software-based correction approach, rather than relying on either estimate alone.

This document is a working draft. The rolling-shutter physics and the two-phase experimental plan are defined; the synthetic validation, the physical rotation rig, and the resulting data are pending.

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Introduction

The LINKU camera system report and the underlying camera configuration analysis did not originally account for spacecraft rotation during image capture. This was identified as a gap: the Arducam HQ 477 (Sony IMX477) sensors used by the system are rolling-shutter sensors, meaning the image is not captured all at once but scanned row by row over a finite readout time. If the spacecraft rotates during that readout window, the resulting image is geometrically sheared, with each row effectively taken from a slightly different attitude.

This is a more serious problem for LINKU than for typical photography, for a reason specific to this mission: the tether itself is only expected to occupy one to three pixels of width in the image, per the sub-pixel observability analysis in the Camera System Report. A rolling-shutter shear of even two or three pixels, which would be a barely noticeable artifact in an ordinary photograph, is comparable in magnitude to the entire object being measured.

A first estimate, worked out from the camera system's own operating parameters and the IMX477 datasheet, is presented in Chapter 1 and indicates that a spacecraft rotation rate at or below approximately 1.05 degree per second (under the software's current 15 fps configuration) to approximately 2.76 degree per second (if a faster sensor readout mode is used) keeps this shear within a tolerable range. This document plans the experiment required to test these estimates directly, rather than relying on them as an assumption, and to validate a software-based correction approach as the near-term mitigation strategy, before considering a hardware change such as a global-shutter sensor.

This document is organized as follows:

- Chapter 1 presents the rolling-shutter physics, the first-pass tolerance estimate, and the relevant correction literature.
- Chapter 2 defines the two-phase experimental methodology: a synthetic validation phase followed by a physical rotation-rig validation phase.
- Chapter 3 defines the experimental setup required for each phase.
- Chapter 4 defines how results will be evaluated once the experiments are executed.

This document is a working draft. The physics background (Chapter 1) and methodology (Chapter 2) are complete. The experimental setup (Chapter 3) and results (Chapter 4) are pending execution.

Chapter 1

Rolling-Shutter Background and First-Pass Estimate

1.1 Rolling-shutter distortion mechanism

A rolling-shutter sensor does not expose its full frame simultaneously. Instead, rows are exposed and read out sequentially, so the top row of an image and the bottom row correspond to slightly different instants in time. For a static camera and a static scene this has no visible effect, but if the camera itself rotates during the readout window, each row records the scene from a slightly different orientation, producing a geometric shear or skew across the image rather than a uniform blur.

This is a distinct phenomenon from ordinary motion blur, which is caused by relative motion during a single row's exposure time and occurs with any shutter type, rolling or global. A rolling-shutter sensor can therefore exhibit both effects at once: shear from rotation during the full-frame readout, and blur from motion during each row's individual exposure.

1.2 First-pass tolerance estimate

Using the camera system's real operating parameters, an order-of-magnitude estimate can be made without any new hardware or experiment.

The instantaneous field of view (IFOV) per output pixel, using the 8 mm focal length lens and the 1920×1440 capture resolution configured in the software, is approximately:

$$\text{IFOV} \approx \frac{\text{pixel pitch (effective)}}{\text{focal length}} \approx 0.023^\circ \text{ per pixel} \quad (1.1)$$

A first, naive bound can be derived from the camera configuration's current software framerate setting of 15 fps, which implies a full-frame readout time on the order of:

$$T_{\text{readout,software}} \approx \frac{1}{15 \text{ fps}} \approx 67 \text{ ms} \quad (1.2)$$

However, this framerate is a conservative software/driver setting, not a sensor limitation. The Sony IMX477 datasheet's Features section states the sensor supports full-resolution readout at up to 60 frame/s (10-bit Normal mode) or 40 frame/s (12-bit Normal mode) [1]. This gives a datasheet-sourced upper bound on the full-frame period of:

$$T_{\text{readout,12-bit}} \approx \frac{1}{40 \text{ fps}} \approx 25 \text{ ms}, \quad T_{\text{readout,10-bit}} \approx \frac{1}{60 \text{ fps}} \approx 16.7 \text{ ms} \quad (1.3)$$

These are still upper bounds: the complete datasheet’s AC Characteristics section (Section 8-2) documents the master clock, PLL, and serial communication timing in detail, but the exact row-scan-only readout time as a function of operating mode is stated to require the separate Software Reference Manual, which is not publicly available [1]. The frame period therefore still includes an unknown amount of blanking time in addition to the actual row-scan time, and the true row-scan-only readout time is expected to be somewhat shorter than either value above.

If a rolling-shutter shear of up to approximately 3 pixels is treated as the tolerable limit, a threshold chosen because the tether itself is only 1 to 3 pixels wide, so any larger shear would be comparable to or larger than the target itself, then the maximum tolerable angular velocity for each readout-time bound is:

$$\omega_{\text{max,software 15fps}} \approx \frac{3 \times 0.023^\circ}{0.067 \text{ s}} \approx 1.05^\circ/\text{s}, \quad \omega_{\text{max,12-bit sensor limit}} \approx \frac{3 \times 0.023^\circ}{0.025 \text{ s}} \approx 2.76^\circ/\text{s} \quad (1.4)$$

This is a meaningful and somewhat unexpected finding: the widely used guidance of keeping spacecraft rotation at or below approximately 1 degree per second appears to reflect the software’s current, conservative 15 fps configuration rather than a hard sensor limitation. If the capture pipeline is reconfigured to use a faster full-resolution readout mode, the sensor itself may tolerate rotation rates 2 to 4 times higher before reaching the same 3-pixel shear limit. This is exactly the kind of estimate the experiment defined in this document is meant to test directly rather than rely on: it depends on an approximated readout time (bounded, not exact) and an assumed tolerable shear threshold, both stated as such above, and it has direct consequences for the ADCS attitude-control-rate requirement recommended in Chapter 4.

1.3 Power cost of faster readout

Running the sensor at a faster readout mode to gain rotation tolerance is not free: it draws more power. The IMX477 datasheet reports the sensor’s current consumption at full resolution, 30 frame/s, as approximately 25.1 mA (typical) to 34.2 mA (maximum) on the analog supply (2.8 V) and 185.0 mA (typical) to 267.5 mA (maximum) on the digital supply (1.05 V), with a small additional interface-supply contribution of 2.4 mA to 2.8 mA at 1.8 V [1]. Combining supply voltages and currents gives an approximate power draw, per camera, of:

$$P_{\text{camera, 30fps, typ}} \approx (2.8\text{V} \times 25.1\text{mA}) + (1.05\text{V} \times 185.0\text{mA}) + (1.8\text{V} \times 2.4\text{mA}) \approx 269 \text{ mW} \quad (1.5)$$

$$P_{\text{camera, 30fps, max}} \approx (2.8\text{V} \times 34.2\text{mA}) + (1.05\text{V} \times 267.5\text{mA}) + (1.8\text{V} \times 2.8\text{mA}) \approx 382 \text{ mW} \quad (1.6)$$

This figure is specified at 30 frame/s, not at the 40 to 60 frame/s modes discussed above; the datasheet does not publish current consumption at those higher rates in the sections available for this document, so the power draw at 40 to 60 fps is expected to be somewhat higher than the values above, not lower.

This is directly relevant to the operational parameters tracked in the Camera System Report (power consumption per stage, Chapter 7): approximately 270 to 380 mW per camera at 30 fps is now a concrete, datasheet-sourced figure for the image sensor itself during active capture, with two LINKU cameras per stereo pair implying roughly 540 to 760 mW combined for the sensors alone, before adding the Raspberry Pi Compute Module 5’s own processing power draw. This closes part of that pending item; the CM5 processing-side power budget remains open.

1.4 Correction approaches in the literature

Two complementary correction strategies are relevant to this application.

Pending: obtain the IMX477 Software Reference Manual, or a bench measurement, to replace this datasheet-feature-list bound with the exact row-scan readout time.

Pending: obtain current-consumption figures at 40 fps and 60 fps specifically (or measure them on the bench) to quantify the exact power

1.4.1 Geometric correction using known angular rate

Karpenko et al. demonstrated that rolling-shutter shear and camera motion can be jointly modeled and corrected using synchronized gyroscope measurements, without requiring machine learning or training data: the row-dependent warp is computed directly from the known angular velocity at the time each row was captured, then geometrically inverted [2]. This is the most directly applicable approach for LINKU, because the ADCS subsystem is expected to provide angular rate telemetry, giving the correction algorithm exactly the input this method requires, and because it does not depend on a training dataset that does not yet exist for this application.

1.4.2 Spacecraft-specific applications

Rolling-shutter correction has also been studied specifically for spacecraft attitude sensors, where distortion from vehicle rotation directly degrades attitude determination accuracy [3]. This confirms that the problem is a recognized concern in flight hardware generally, not one specific to low-cost COTS sensors, although the correction requirements for a star sensor (arc-second attitude accuracy) are substantially stricter than what is required here.

1.4.3 Learning-based correction

More recent work has explored learning-based rolling-shutter correction assisted by IMU data [4]. This approach is noted for completeness but is not adopted as the primary method for this document, since it requires a training dataset of matched rolling-shutter and global-shutter image pairs that does not currently exist for this application, and the geometric approach of Section 1.2 is expected to be sufficient given that the correction only needs to remove a known, physically modeled shear rather than recover general motion blur.

This specific citation could not be freely downloaded (IEEE Xplore paywall, no open-access copy found) and its author list is unverified. Treat this paragraph's claim as plausible but unconfirmed until a full copy is obtained, for example through institutional IEEE Xplore access.

Chapter 2

Experimental Methodology

2.1 Two-phase strategy

Validating a rolling-shutter correction approach requires two things that are not both available immediately: a known, controllable angular velocity to test against, and a way to tell whether the correction actually worked. This document splits the validation into two phases so that the correction algorithm itself can be tested before any new hardware needs to be built.

Phase 1: synthetic validation (no new hardware required)



Phase 2: physical validation (requires the rotation rig)



Figure 2.1: Two-phase validation strategy. Phase 1 tests whether the correction algorithm is mathematically sound using a synthetically warped static image. Phase 2 tests the same correction against a real rotating capture, once the rig exists.

2.1.1 Phase 1: synthetic validation

A static, already-captured laboratory tether photograph is used as ground truth. A synthetic rolling-shutter shear is applied to it, computed from a chosen angular velocity and the readout-time model of Section 1.2, producing an image equivalent to what the camera would have captured while rotating at that rate. The correction algorithm is then applied using that same known angular velocity as input, and its output is compared back against the original static image.

This phase isolates the correction algorithm from any hardware uncertainty: the ground truth, the applied distortion, and the angular velocity are all exactly known by construction, so any residual error after correction is attributable to the algorithm itself rather than to measurement uncertainty.

2.1.2 Phase 2: physical validation

The camera and tether proxy are mounted so that the camera assembly rotates at a controlled, known angular velocity while imaging the tether inside the dark enclosure. The same correction algorithm is applied using the rig's commanded or measured angular velocity, standing in for the angular-rate telemetry that would be provided by the ADCS subsystem in flight. Results are compared against a static baseline capture of the same tether configuration.

This phase validates the full chain, including effects that Phase 1 cannot capture, such as illumination changes during rotation, vibration from the rig itself, and any timing mismatch between the commanded angular velocity and the camera’s actual exposure window.

2.2 Sweep of angular velocities

Both phases should test a sweep of angular velocities rather than a single value, to characterize how performance degrades with increasing rotation rate rather than only confirming or denying a single threshold:

0, 0.25, 0.5, 1.0, 2.0, 5.0 degrees per second

The 1.0 and 2.0 degree per second points are included specifically to bracket the two first-pass estimates of Section 1.2 (approximately 1.05 degree per second under the current 15 fps software configuration, and approximately 2.76 degree per second under the sensor’s faster full-resolution readout modes). Points below and above these characterize the margin around both estimates.

2.3 Evaluation metrics

For each angular velocity tested, in both phases, the following are computed and compared against the static baseline, reusing the project’s existing reconstruction pipeline rather than introducing new metrics:

- SmartWireTracker coverage percentage.
- Endpoint detection position error.
- Traced path length error.
- 3D reconstruction curvature and straightness error, where a stereo pair is available.

Each metric is reported for the uncorrected distorted image and for the corrected image, at each angular velocity, so that the correction’s benefit (the difference between the two) is explicit rather than only the corrected result in isolation.

Chapter 3

Experimental Setup

This chapter defines what needs to be built to execute the methodology of Chapter 2. All items in this chapter are pending; none have been executed yet.

3.1 Phase 1 setup: synthetic validation

Pending: implement the row-dependent shear model (Section 1.2) as a function that takes a static image and an angular velocity and returns the synthetically sheared image. Implement the corresponding inverse (correction) function that takes a sheared image and the same angular velocity and returns the corrected image. Select 3 to 5 representative static tether photographs already available in the existing dataset to use as ground truth for this phase, covering straight, curved, and if available a self-crossing configuration.

3.2 Phase 2 setup: physical rotation rig

Pending: design and build a rotation stage (stepper motor or servo-driven turntable) capable of controlled, known angular velocities in the range defined in Chapter 2, with either commanded-rate accuracy or an encoder providing measured rate, since the correction algorithm's input quality directly determines its output quality. Mount the stereo camera rig on this stage such that it rotates while imaging the tether proxy inside the dark enclosure, following the same illumination and enclosure configuration used for the Camera System Report experiments.

Pending: define how the rig's angular velocity will be communicated to the correction algorithm for each capture (for example, a logged commanded value, or an encoder reading synchronized to the capture timestamp), since this stands in for the ADCS angular-rate telemetry that would be available in flight.

3.3 Software implementation

Pending: implement the correction as a processing step compatible with the existing pipeline, positioned before segmentation, so that its output can be fed into the unmodified downstream stages (segmentation, endpoint detection, path tracing) for the comparison defined in Chapter 2.

Chapter 4

Evaluation and Reporting Plan

This chapter defines how results will be evaluated and reported once the experiments of Chapter 3 are executed. No results exist yet; this chapter defines the criteria that will be applied to them.

4.1 Reporting format

For each angular velocity tested, in each phase, the following will be reported:

- The uncorrected distorted image and the corrected image, for visual reference.
- The four evaluation metrics of Chapter 2, for the uncorrected and corrected image, against the static baseline.
- A plot of each metric's degradation versus angular velocity, with and without correction, to visualize where performance drops below an acceptable level.

4.2 Interpretation criteria

Pending decision, to be set before running the experiment: define the maximum acceptable degradation in each metric (for example, endpoint position error, path coverage percentage) relative to the static baseline, below which imagery at a given angular velocity is considered usable. This threshold should be consistent with the operational tolerances already used elsewhere in the Camera System Report, rather than chosen independently.

4.3 Expected outcome and operational recommendation

The intended output of this experiment is a single operational number: the maximum spacecraft rotation rate, with correction applied, at which tether imagery remains usable by the existing pipeline. This number will either confirm the first-pass estimate of approximately 1 degree per second from Section 1.2, or replace it with an experimentally validated value, which is the primary purpose of this document.

This recommendation is intended to be provided to the ADCS subsystem as an attitude-control-rate requirement during tether imaging windows, and should be read together with the operational parameters (power consumption per stage, image acquisition rate) tracked separately in the Camera System Report, Chapter 7, since attitude control, power, and imaging cadence are jointly constrained by the same imaging windows.

Chapter 5

Conclusions and Status

This document formalizes the rotation and rolling-shutter mitigation plan that was identified as a gap in the original Camera System Report. A first-pass, parameter-based estimate places the maximum tolerable spacecraft rotation rate between approximately 1.05 degree per second (under the software's current 15 fps configuration) and approximately 2.76 degree per second (if the sensor's faster full-resolution readout modes are used, per the IMX477 datasheet), and a two-phase experimental plan is defined to test these estimates directly and to validate a geometric, gyroscope-rate-based correction approach following the precedent established by Karpenko et al. [2].

The software-based correction approach was selected as the near-term direction over a hardware change such as a global-shutter sensor, since it can be validated using the existing camera hardware and does not require a new procurement cycle. A global-shutter sensor remains a viable longer-term option if the software correction proves insufficient at the rotation rates required by the mission's attitude control budget.

5.1 Current status

- Rolling-shutter physics and first-pass estimate: complete.
- Literature review and correction approach selection: complete.
- Two-phase methodology definition: complete.
- Phase 1 (synthetic validation) implementation: pending.
- Phase 2 (physical rotation rig) design and construction: pending.
- Results and operational recommendation: pending.

5.2 Next steps

1. Obtain the IMX477 datasheet readout timing to replace the framerate-based approximation used in Section 1.2.
2. Implement and run the Phase 1 synthetic validation.
3. Design and build the Phase 2 rotation rig.
4. Run the Phase 2 physical validation and compare against Phase 1.
5. Report the validated maximum operational rotation rate to the ADCS subsystem as an attitude-control-rate requirement during tether imaging windows.

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